Notes on interpreting the estimation error

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1 Introduction

Jeon et al. [JLLVR24] derive that the optimal log-loss satisfies

$$\mathbb{L}_T = \frac{\mathbb{H}(H_T|\theta)}{T} + \frac{\mathbb{I}(H_T;\theta)}{T},$$

naming the two terms *irreducible error* and *estimation error* respectively. In the meta-learning setting they further decompose

$$\mathbb{L}_{M,T} = \frac{\mathbb{H}(H_{M,T}|\theta_{1:M})}{MT} + \frac{\mathbb{I}(H_{M,T};\psi)}{MT} + \frac{\mathbb{I}(D_m;\theta_m|\psi)}{T}.$$

The implicit suggestion is that these decompositions carry intrinsic meaning: the irreducible error reflects aleatoric uncertainty, and the estimation errors reflect statistical learning difficulty. In this document we take note with regard to the interpretability of this decomposition.

First, the single-sequence decomposition is not canonical. For any random sequence $\{X_t\}$ and any $\alpha \in [0,1]$, there exists a parameterization Θ_α such that $\{X_t\} \sim A(\Theta_\alpha)$ and

$$\frac{\mathbb{I}(H_T;\Theta_\alpha)}{T} = \alpha \cdot \frac{\mathbb{H}(H_T)}{T}.$$

That is, the estimation error can be set to any fraction of total entropy by choice of parameterization, with the irreducible error absorbing the remainder.

Second, the meta-learning decomposition places no restriction on the (D_m, θ_m) factorization sufficient to pin down the irreducible term. In particular, setting $\theta_m = D_m$ yields $\mathbb{H}(D_m|\theta_m,\psi) = 0$, collapsing the irreducible error entirely and pushing all uncertainty into the estimation terms. The labeling of any term as "irreducible" is therefore a consequence of the chosen parameterization, not of the sequence itself.[1]

We construct the two extremal cases of the first critique explicitly (Lemmas 3.1, 3.2) and interpolate between them (Theorem 3.4).

2 Initial Definitions

2.1 General definitions

Definition 2.1. An alphabet is a finite non-empty set of tokens.

Definition 2.2. A word over an alphabet is a tuple containing just tokens of that alphabet.

[1] There could be a canonical factorization of (D_m, θ_m), such that with de Finetti's theorem all θ_m obeying this factorization would yield the same value for $\mathbb{H}[D_m|\theta]$.

Definition 2.3. Σ^∞ is the set of infinite words over the alphabet Σ.

Definition 2.4. Σ^* is the set of finite words over the alphabet Σ.

Definition 2.5. Σ^n is the set of words over the alphabet Σ of length n.

Definition 2.6. By $X \sim \mathbb{R}^n$, we denote that X is a random vector in $\mathbb{R}^n$.

2.2 Definitions related to this script

Definition 2.7. $\mathcal{P}_\Sigma$ is the set of probability distribution functions of a random variable that takes a value of alphabet Σ. More formaly $\mathcal{P}_\Sigma = \{f \in \{\Sigma \to [0,1]\} | \sum_{i\in\Sigma} f(i) = 1\}$.

Definition 2.8. $\mathcal{P}^*_\Sigma$ is the set of functions that, take in a word w over the alphabet Σ and return a probability distribution function of a random variable that takes a value of Σ. $\mathcal{P}^*_\Sigma := \{\Sigma^* \to \mathcal{P}_\Sigma\}$. An element of P^*_Σ is called a parametrized autoregressive model (PAM).

Remark 2.1. Why we call a parametrized autoregressive model such will become, more clear after defining autoregressive models.

Definition 2.9. Let $\{X_t\}_{t\in\mathbb{N}}$ be a sequence, then

- $X_{i:j} := (X_i, X_{i+1}, ..., X_j)$
- $H_T := X_{1:T}$
- $H_0 := ()$

Definition 2.10. BACKUO SSSHIT: An autoregressive sequence $\{X_t\}_{t\in\mathbb{N}}$ is called an autoregressive sequence on $\mathcal{P} \in \mathcal{P}^*_\Sigma$ if $\forall_{T\in\mathbb{N}}.\mathbb{P}[H_{T-1} = (x_1, ..., x_{T-1})] > 0 \implies \mathbb{P}[X_T = x_T | H_{T-1} = (x_1, ..., x_{T-1})] = P(x_1, ..., x_{T-1})(x_T)$. We denote this by $\{X_t\}_{t\in\mathbb{N}} \sim \mathcal{P}$.

Definition 2.11. We call a measurable map $A : \mathbf{\Theta} \to \mathcal{P}^*_\Sigma$ be a measurable map an Autoregressive Model (AM).

Definition 2.12 (Bayesian autoregressive sequence)**.** Let $A : \mathbf{\Theta} \to \mathcal{P}^*_\Sigma$ be an AM and let Θ be a random variable taking values in $\mathbf{\Theta}$.

We call $\{X_t\}_{t\in\mathbb{N}}$ a Bayesian autoregressive sequence with representation (A, Θ) if for all $t \in \mathbb{N}$ and all $x \in \Sigma$:

$$\mathbb{P}(X_{t+1} = x \mid H_t, \Theta) = A(\Theta)(H_t)(x) \quad \text{a.s.}$$

We write

$$\{X_t\}_{t\in\mathbb{N}} \sim A(\Theta).$$

Remark 2.2. Note that for every random sequence $\{X_t\}_{t\in\mathbb{N}}$ over an alphabet, there is a PAM $\mathcal{P}$ representing $\{X_t\}_{t\in\mathbb{N}}$.

Furthermore if $f, g \in \mathcal{P}^*_\Sigma$ and $\{X_t\}_{t\in\mathbb{N}} \sim f$ and $\{Y_t\}_{t\in\mathbb{N}} \sim g$, have different distributions, then $f \neq g$.

The reverse scenario is not necessarily true. Consider a case where $\forall_T \mathbb{P}[H_T = w] = 0$. Then both PAMs with $f(w)(i) = 1$ and $g(w)(i) = .5$ could represent $\{X_t\}_{t\in\mathbb{N}}$.

Remark 2.3. Be aware: other people use the word Autoregressive Model for Parametrized Autoregressive Models and some other people use the word Autoregressive Model for something even more different.

Let's look at what is an AM A. I hope I didn't make it look more complicated than it is. An AM is a function that - by definition - maps parameters to PAMs. A PAM is nothing more than an expression of a probability distribution of a sequence of tokens. For example, a transformer architecture can be seen as an AM and if you give the parameters of this architecture values, then you obtain a transformer, which can be seen as a PAM.

To be more specific: Let A be an AM, then $A(\theta)$ is a PAM, $A(\theta)(x_1, ..., x_{t-1})$ represents a probability distribution function for a random variable that takes an element of the alphabet and $A(\theta)(x_1, ..., x_{t-1})(x_t)$ represents a probability.

2.3 Definitions and results from [JLLVR24]

Definition 2.13. Let π denote an estimator, for a probability distribution of the next variable X_{t+1} given the previous random sequence of tokens H_t. π can be dependent on the Bayesian prior distribution but not on the Bayesian prior itself.

Definition 2.14. The average cumulative expected log-loss: $\mathbb{L}_T(\pi) := \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[-\log(\pi(H_t, X_{t+1}))]$

Definition 2.15. $\mathbb{L}_T := \min_\pi \mathbb{L}_T(\pi)$

Theorem 2.1. *[JLLVR24]* $\mathbb{L}_T T = \mathbb{I}(H_T;\theta) + \mathbb{H}(H_T|\theta)$

Definition 2.16. $\frac{\mathbb{I}(H_T;\theta)}{T}$ is called the estimation error.

Definition 2.17. $\frac{\mathbb{H}(H_T|\theta)}{T}$ is called the irreducible error.

3 Results

Lemma 3.1. *There is an AM A_0 over Σ, such that for every random sequence $\{X_t\}_{t\in\mathbb{N}}$ on Σ, there is $\Theta_0 \in \mathbb{R}^n$, such that $\{X_t\}_{t\in\mathbb{N}} \sim A_0(\Theta_0)$, with the estimation error $\mathbb{I}(H_T;\Theta_0)/T = 0$.*

Idea of the proof: We can easily show the existence of an AM A that can represent any random sequence on Σ, with deterministic parameters. The lemma follows immediately, as the determenistic parameters have the entropy of zero.

Proof.

1. To show the existence of a surjective AM it suffices to show that $|\mathbb{R}^n| \geq |\mathcal{P}^*_\Sigma|$.

$$|\mathcal{P}^*_\Sigma| \leq |\{\Sigma^* \to \{\Sigma \to [0,1]\}\}| = |\{\Sigma^* \to \{\Sigma \to \mathbb{R}\}\}| = |\{\Sigma^* \to \mathbb{R}\}| \leq |\{\mathbb{N} \to \mathbb{R}\}| = |\mathbb{R}|$$

2. Let A_0 be a surjective AM.
3. Consider any random sequence $\{X_t\}_{t\in\mathbb{N}}$ on Σ
4. Consider this function $f : \Sigma^* \to \mathcal{P}_\Sigma$:

$$f : w \mapsto (x \mapsto \mathbb{P}[X_{|w|} = x | H_{|w|-1} = w]) \tag{1}$$

   5. $f \in \mathcal{P}^*_\Sigma$ (f is a PAM). (By definition of f)
   6. $\{X_t\}_{t\in\mathbb{N}} \sim f$. (By construction of f and by definition 2.12)
7. $\exists_{\Theta_0\in\mathbb{R}^n}.A_0(\Theta_0) = f$. (From A_0 surjective)
8. $\{X_t\}_{t\in\mathbb{N}} \sim A_0(\Theta_0)$
9. If we model $\{X_t\}_{t\in\mathbb{N}} \sim A_0(\Theta_0)$, then $\mathbb{H}(\Theta_0) = 0$ (since $\Theta_0 \in \mathbb{R}^n$) and therefore we obtain an estimation error of 0: $\mathbb{I}(H_T;\Theta_0) = 0$.

□

Definition 3.1. We call an AM A_0, that satisfies the condition of lemma 3.1, a distributionaly full AM.

Lemma 3.2. *There is an AM A_1 over Σ, such that for every random sequence $\{X_t\}_{t\in\mathbb{N}}$ on Σ, there is $\Theta_1 \sim R_n$, such that $\{X_t\}_{t\in\mathbb{N}} \sim A_1(\Theta_1)$, with the estimation error $\mathbb{I}(H_T;\Theta_1)/T = \mathbb{H}(H_T)/T$.*

Idea of the proof: We let Θ_1 suck up all randomness by $\Theta_1 := \{X_t\}_{t\in\mathbb{N}}$ and the rest is just formal stuff.

Proof.

1. We define $A_1 : \Sigma^\infty \to \mathcal{P}^*_\Sigma$ by:

$$A_1(\theta) := h \mapsto \big(x \mapsto \mathbb{1}[\theta_{|h|+1} = x]\big)$$

2. Let $\Theta_1 := \{X_t\}_{t\in\mathbb{N}}$.
3. We show $\{X_t\}_{t\in\mathbb{N}} \sim A_1(\Theta_1)$.

4. By definition, it suffices to verify that for all $t \in \mathbb{N}$ and $x \in \Sigma$:

$$\mathbb{P}(X_{t+1} = x \mid H_t, \Theta_1) = A_1(\Theta_1)(H_t)(x) \quad \text{a.s.}$$

5. Since $\Theta_1 = \{X_t\}_{t\in\mathbb{N}}$, we have:

$$\mathbb{P}(X_{t+1} = x \mid H_t, \Theta_1) = \mathbb{P}(X_{t+1} = x \mid H_t, \{X_t\}_{t\in\mathbb{N}}).$$

6. Now X_{t+1} is measurable with respect to $\sigma(\{X_t\}_{t\in\mathbb{N}})$, hence:

$$\mathbb{P}(X_{t+1} = x \mid H_t, \{X_t\}_{t\in\mathbb{N}}) = \mathbb{1}[X_{t+1} = x] \quad \text{a.s.}$$

7. On the other hand, by definition of A_1:

$$A_1(\Theta_1)(H_t)(x) = \mathbb{1}[\Theta_{1,t+1} = x] = \mathbb{1}[X_{t+1} = x].$$

8. Therefore:

$$\mathbb{P}(X_{t+1} = x \mid H_t, \Theta_1) = A_1(\Theta_1)(H_t)(x) \quad \text{a.s.}$$

9. This proves $\{X_t\}_{t\in\mathbb{N}} \sim A_1(\Theta_1)$.

10. $\mathbb{H}[H_T|\Theta_1] = \mathbb{H}[\Theta_1|\Theta_1] = 0.$ Def. Θ_1

11. $\mathbb{H}(H_T)/T = \mathbb{H}[H_T|\Theta_1]/T + \mathbb{I}(H_T;\Theta_1)/T = \mathbb{I}(H_T;\Theta_1)/T$

□

Definition 3.2. We call an AM A_1, which satisfies the condition of Lemma 3.2, a deterministically full AM.

Lemma 3.3.

- *Let X be a discrete random variable, Y be a random variable and A an random event, (such that $\mathbb{P}[A] \in (0,1)$.*
- *(I:) Let for all $x \in \mathcal{X}, y \in \mathcal{Y}$ hold that $f_{Y|A}(y) > 0 \implies \mathbb{P}[X = x|Y = y] = \mathbb{P}[X = x|Y = y, A]$.*
- *(II:) Let for all $x \in \mathcal{X}, y \in \mathcal{Y}$ hold that $f_{Y|\overline{A}}(y) > 0 \implies \mathbb{P}[X = x|Y = y] = \mathbb{P}[X = x|Y = y, \overline{A}]$.*

Then $\mathbb{H}[X|Y] = \mathbb{P}[A]\mathbb{H}[X|Y, A] + (1 - \mathbb{P}[A])\mathbb{H}[X|Y, \overline{A}]$

Proof.

1. $\mathbb{H}[X|Y] = \mathbb{E}[y \mapsto \mathbb{H}[X, Y = y](Y)]$
2. $$= \mathbb{P}[A]\mathbb{E}[y \mapsto \mathbb{H}[X, Y = y](Y)|A] + (1 - \mathbb{P}[A])\mathbb{E}[y \mapsto \mathbb{H}[X, Y = y](Y)|\overline{A}]$$
   (from law of total expectation)
3. We examine $\mathbb{E}[y \mapsto \mathbb{H}[X, Y = y](Y)|J]$ for the cases $J = A$ and $J = \overline{A}$
   4. $\mathbb{E}[y \mapsto \mathbb{H}[X, Y = y](Y)|J]$ (value is defined since $\mathbb{P}[J] > 0$)

5. $$= \int_{\mathcal{Y}} f_{Y|J}(y)\mathbb{E}[x \mapsto -\log(\mathbb{P}[X = x|Y = y]|Y)(X)]\, dy$$

(by writing out the expectation and the entropy)

6. $$= \int_{\mathcal{Y}} f_{Y|J}(y)\mathbb{E}[x \to -\log(\mathbb{P}[X = x|Y = y, J]|Y)(X)]\, dy$$

(by I if $J = A$ and by II if $J = \overline{A}$)

7. $$= \int_{\mathcal{Y}} f_{Y|J}(y) \sum_{x\in\mathcal{X}} [-\log(\mathbb{P}[X = x|Y = y, J])\mathbb{P}[X = x|Y = y]]\, dy$$

8. $$= \int_{\mathcal{Y}} f_{Y|J}(y) \sum_{x\in\mathcal{X}} [-\log(\mathbb{P}[X = x|Y = y, A])\mathbb{P}[X = x|Y = y, J]]\, dy$$

(by I if $J = A$ and by II if $J = \overline{A}$)

9. $$= \int_{\mathcal{Y}} f_{Y|J}(y)\mathbb{H}[X = x|Y = y, J]\, dy$$

10. $= \mathbb{H}[X|Y, J]$

11. We can now insert the reult from 4 to 10 into 2

12. $= \mathbb{P}[A] \cdot \mathbb{H}[X = x|Y, A] + (1 - \mathbb{P}[A]) \cdot \mathbb{H}[X = x|Y, \overline{A}]$

□

Comments on this Lemma: Possibly the condition in the lemma could me much shorter and more intuitive. This lemma could require less text to apply if one shows $I \implies II$ (I attached a proof sketch to the end to the document (dont read it, it is messy). On request I can formalize it to see if it is actually true). Also it might be possible that I and II are almost equivalent with X and A are independent given Y. I put the condition of the lemma like this, because I think in total we require less text by doing so. Also where we are going to use the lemma, the condition will be easy to check. But it might also be possible, that everything looks cleaner if we use independence.

Theorem 3.4. *There is an AM A_c, such that given any $\alpha \in [0, 1]$ and any random sequence $\{X_t\}_{t\in\mathbb{N}}$ there is $\Theta_\alpha \sim \mathbb{R}^n$, such that $\{X_t\}_{t\in\mathbb{N}} \sim A_c(\Theta_\alpha)$, with the estiamtion error $\mathbb{I}(H_T; \Theta_1)/T = \alpha \cdot \mathbb{H}(H_T)/T$.*

Idea of the proof: We know there is A_0 and A_1, such that of any sequence $\{X_t\}_{t\in\mathbb{N}}$ on the same alphabet, there is Θ_0 and Θ_1, such that $X \sim A_0(\Theta_0)$ with minimal estimation error and $X \sim A_1(\Theta_1)$ with maximal estimation error. Let us suppose a random variable Θ_c that is with some probability a Θ_1 and otherwise Θ_0 and, which also "stores" which of the two variables it is. We combine A_0 and A_1 into an AM A_c, such that A_c is an AM that "chooses" whether to use A_0 or A_1, depending on $\Theta_\alpha = \Theta_0$ or $\Theta_\alpha = \Theta_1$. Then $X \sim A_c(\Theta_c)$. All that is left is to see whether we can choose a such that $\mathbb{I}(H_T; \Theta_c) = \alpha \cdot \mathbb{H}(H_T)$. (Spoiler alert: $a = \alpha$)

Proof.

1. Let A_0 be a distributionaly full AM and let A_1 a deterministicaly full AM.
2. Let $\Theta_0 \in \mathbb{R}^n$, such that $X \sim A_0(\Theta_0)$ and $\mathbb{I}(H_T; \Theta_0) = 0$.
3. Θ_0 exists (from Lemma 3.1)

4. Let $\Theta_1 \sim \mathbb{R}^n$, such that $X \sim A_1(\Theta_1)$ and $\mathbb{I}(H_T; \Theta_1) = \mathbb{H}(H_T)$.
    5. Θ_1 exists (from Lemma 3.2)
6. From the existance of Θ_0, A_0 and Θ_1, A_1, we can assume w.l.o.g. that $\alpha \in (0, 1)$.
7. Let B be a binary random variable with $\mathbb{P}(B = 1) = \alpha$ and $\mathbb{P}(B = 0) = 1 - \alpha$.
8. Let $\Theta_\alpha \sim \mathbb{R}^{n+1}$, with $(\Theta_\alpha)_{n+1} := B$ and $(\Theta_\alpha)_{1:n} := (1 - B) \cdot \Theta_0 + B \cdot \Theta_1$.
9. We define AM $A_c : \mathbb{R}^{n+1} \to \mathcal{P}^*_\Sigma$:

$$A_c(\Theta) := A_0((\Theta)_{0:n}) \cdot (1 - (\Theta)_{n+1}) + A_1((\Theta)_{0:n}) \cdot (\Theta)_{n+1}$$

    10. given $B = 1$: $A_c(\Theta) = A_1((\Theta)_{0:n}) = A_1(\Theta_1) \implies X \sim A_c(\Theta_\alpha)$
    11. given $B = 0$: $A_c(\Theta) = A_0((\Theta)_{0:n}) = A_0(\Theta_0) \implies X \sim A_c(\Theta_\alpha)$
    12. $\mathbb{P}[B \notin \{0, 1\}] = 0$
    13. $X \sim A_c(\Theta_\alpha)$ (from 10, 11, 12)
14. In the next steps we evaluate: $\mathbb{I}(H_T; \Theta_\alpha) = \mathbb{H}[H_T] - \mathbb{H}[H_T|\Theta_\alpha]$
    15. We simplify $\mathbb{H}[H_T|\Theta_\alpha]$ using lemma 3.3.
    16. We first show that we can apply this lemma, i.e. that for all $x \in \Sigma^T, y \in \mathbb{R}^{n+1}$ holds that
        17. I: $f_{\Theta_\alpha|B=1}(y) > 0 \implies \mathbb{P}[H_T = x|\Theta_\alpha = y] = \mathbb{P}[H_T = x|\Theta_\alpha = y, B = 1]$.
        18. II: $f_{\Theta_\alpha|B=0}(y) > 0 \implies \mathbb{P}[H_T = x|\Theta_\alpha = y] = \mathbb{P}[H_T = x|\Theta_\alpha = y, B = 0]$.
    19. I and II are true, since $B = (\Theta_\alpha)_{n+1}$
    20. Now we apply lemma 3.3:

$$\mathbb{H}[H_T|\Theta_\alpha] = \alpha\mathbb{H}[H_T|\Theta_\alpha, B = 1] + (1 - \alpha)\mathbb{H}[H_T|\Theta_\alpha, B = 0]$$

    21. $= \alpha\mathbb{H}[H_T|\Theta_1] + (1 - \alpha)\mathbb{H}[H_T|\Theta_0]$ (Def. Θ_α)
    22. $= \alpha \cdot 0 + (1 - \alpha)\mathbb{H}[H_T] = (1 - \alpha)\mathbb{H}[H_T]$ (Def. Θ_0, Θ_1)
23. $\mathbb{I}(H_T; \Theta_\alpha) = \mathbb{H}[H_T] - \mathbb{H}[H_T|\Theta_\alpha] = \alpha\mathbb{H}[H_T]$ (From 15 - 22)
24. $\mathbb{I}(H_T; \Theta_\alpha)/T = \alpha\mathbb{H}[H_T]/T$

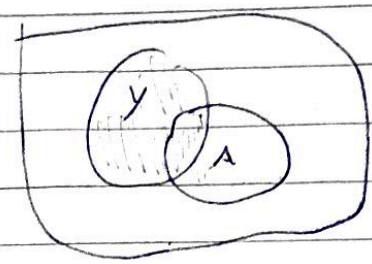
□

References

[JLLVR24] Hong Jun Jeon, Jason D Lee, Qi Lei, and Benjamin Van Roy. An information-theoretic analysis of in-context learning. *arXiv preprint arXiv:2401.15530*, 2024.

4 Below Proof Sketch, that I imples II in lemma 3.3

$f_{Y|A}(y) > 0$



- Case $f_{Y|A}(y), f_{Y|\bar{A}}(y) > 0$:

$$P[X=x \mid Y=y] = P(X=x \mid Y=y, A]$$

$$\Rightarrow P(X=x \mid Y=y, \bar{A}] = P(X=x \mid Y=y)$$

bayes.

- Case $f_{Y|A}(y) = 0$, $f_{Y|\bar{A}}(y) > 0$:

Assume

$$P[X=x \mid Y=y] \neq P(X=x \mid Y=y, \bar{A}]$$

since $f_{Y|A}(y) = 0$; $f_Y(y) > 0$ $(f_{Y|\bar{A}}(y) > 0 \Rightarrow f_Y(y) > 0)$

$$\Rightarrow \mathbb{P}(\bar{A} \mid Y=y) = 1 \Rightarrow \mathbb{P}(\bar{A} \mid X=x, Y=y) = 1$$

~~by the law of total~~ def. cond. f

$$\cancel{P(X=x \mid Y=y, \bar{A}) = P(Y=}$$

$$\cancel{P(X=x \mid Y=y) = \mathbb{P}(\bar{A})\, P(X=x \mid Y=y, \bar{A})\, \mathbb{P}(\bar{A} \mid X=x)}$$

- bayes :

$$P(X=x \mid Y=y, \bar{A}] = \overbrace{P(\bar{A} \mid Y=y, X=x]}^{1} \underbrace{\frac{P(X=x \mid Y=x)}{P(\bar{A} \mid Y=y)}}_{1}$$

$\Rightarrow$ contradict assumption $\square$